

COURSE TITLE : LATEX TECHNOLOGY
COURSE CODE : 4092
COURSE CATEGORY : B
PERIODS/WEEK : 5
PERIODS/SEMESTER: 70/4
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Compounding ingredients and latex compounding	18
2	Dipped products and latex casting	16
3	Production of latex foam and coir foam	18
4	Production of latex extruded products , adhesives and latex spreading	18
TOTAL		70

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To know about compounding ingredients and latex compounding
2	2	To understand dipped products and latex casting
3	3	To understand the production of latex foam and coir foam
4	4	To understand the production of latex extruded products ,adhesives and latex spreading

SPECIFIC OUTCOME

MODULE I COMPOUNDING INGREDIENTS AND LATEX COMPOUNDING

1.1.0 To know about compounding ingredients and latex compounding

- 1.1.1. Discuss the importance of Natural rubber latex in product manufacturing.
- 1.1.2. Define compounding of latex and state the various compounding ingredients used in latex.
- 1.1.3. Explain the principles of latex compounding.
- 1.1.4. Distinguish between latex compounding and dry rubber compounding.
- 1.1.5. State the functions and dosages of Vulcanizing agents, accelerators , activators, fillers, stabilisers, wetting agents, thickeners, gelling agents, heat sensitizing agents, emulsifiers, dispersing agents and antidegradants.

- 1.1.6. Describe the methods of preparation of ZnO, Sulphur, ZMBT dispersions with typical formulations.
- 1.1.7. Describe the methods of preparation of SP and Oil emulsions.
- 1.1.8. Explain the working and different parts of Ball mill.
- 1.1.9. Describe the characteristics and applications of synthetic latices in latex products manufacture.

MODULE II DIPPED PRODUCTS AND LATEX CASTING

2.1.0 To Understand dipped products and latex casting

- 2.1.1. State the principle of dipping process.
- 2.1.2. Explain the different methods of dipping process with special reference to straight dipping, coagulant dipping and heat sensitised dipping.
- 2.1.3. Give the relation between the thickness of deposit formed and number of dips.
- 2.1.4. Describe the after treatment of latex deposit -drying, surface treatment, vulcanisation and stripping.
- 2.1.5. Explain the principles and method of halogenations. Leaching, jazzing and chlorination.
- 2.1.6. Explain the process of manufacture and characteristics of dipped goods like rubber and, balloons, condoms, gloves, nipples and finger caps with typical formulations.
- 2.1.7. Discuss the common defects and their remedies of dipped goods.
- 2.1.8. State the general principles involved in the production of articles from latex by casting
- 2.1.9. Describe the different steps involved in the production of latex cast articles.
- 2.1.10. Explain the preparation of moulds for latex casting.

MODULE III PRODUCTION OF LATEX FOAM AND COIR FOAM

3.1.0 To understand the production of latex foam and coir foam

- 3.1.1 List the various latex foam products and their manufacturer in India
- 3.1.2. Distinguish Dunlop process and Talalay process
- 3.1.3. Explain the different steps in the production of latex foam by Dunlop process with typical formulations.
- 3.1.4. Define gelatin of latex and give a comparative study of the various gelling systems
- 3.1.5. Describe the various steps involved in the production of coir foam.
- 3.1.6. Explain the different steps in the production of Carpet backing with a typical formulation
- 3.1.7. Give the compositions, mechanism of setting, advantages and applications of latex cement flooring mixture.
- 3.1.8. Explain the method of application of latex in road surfacing with its advantages.
- 3.1.9. Explain the paper treatment methods and its application

MODULE IV PRODUCTION OF LATEX EXTRUDED PRODUCTS, ADHESIVES AND LATEX SPREADING

4.1.0 To understand the production of latex extruded products, adhesives and latex spreading

- 4.1.1. State the principle of Latex extrusion.

- 4.1.2. Explain the method of manufacture of latex thread with details of process, formulation and equipments.
- 4.1.3. Discuss the quality control tests of latex thread.
- 4.1.4. Describe the process and equipment involved in the production of latex rubber tubing '
- 4.1.5. List different types of latex adhesives, compare it with solvent based adhesive
- 4.1.6. Explain the method of preparation of adhesives for paper, leather, metal, ceramics and glass.
- 4.1.7. State the general principles and process of impregnation and spreading of latex.
- 4.1.8. Explain the machinery details used for latex spreading.

CONTENT DETAILS

MODULE-I

Compounding ingredients and latex compounding.

Compounding ingredients –importance of natural rubber latex in product manufacturing. Compounding of latex-principles of latex compounding -difference between latex compounding and dry rubber compounding. Functions and dosages of vulcanizing agents-accelerators-activators –fillers-stabilizers-wetting agents-thickeners-gelling agents-heat sensitizing agents-emulsifiers-dispersing agents-antidegradants. Method of preparation of Zno-sulphur-ZMBT-dispersion with typical formulations-working and parts of ball mill. Method of preparation of SP and oil emulsion. Characteristics and applications -synthetic lattices in latex product manufacture-SBR latex.

MODULE-II

Dipped products and latex casting.

Dipping-principal and procedure of dipping process-different types of dipping –after treatment of latex deposits -Manufacture of dipped goods with formulation and flow chart-defects and remedies . latex casting –principle and procedure of casting-production of cast articles –mould preparation .

MODULE-III

Production of latex foam and coir foam

List latex foam products and their manufactures –method of foam production –Dunlop process and talalay process –different steps –principle of gelation and gelling agents –production of coir foam-different steps –carpet backing –typical formulations. latex cement –preparation-advantages-formulation. Different method of paper treatment and its applications.

MODULE -IV

Production of latex extruded product, adhesives and latex spreading

Latex extrusion-principle –formulation-equipments-process- products-equality control .latex thread, latex tubing. Latex adhesives –formulation-preparation-compare with solvent based adhesives. different types of adhesives . latex spreading – formulation-impregnation-machinery-for latex spreading – different products .

REFERENCE BOOKS

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| 1. High Polymer Lattices, vol-I, vol-II, Vol-III | - | D.C.Blackley (Publisher –Springer Netherlands,1997) |
| 2. Latex Industry | - | R.J.Noble (Publisher – Rubber Age) |
| 3. Latex Manual | - | ICI Handbook |
| 4. Rubbery Material | - | J.Brydson (Publisher -Elsevier Applied Science, London, 1988) |